

2025/12/12

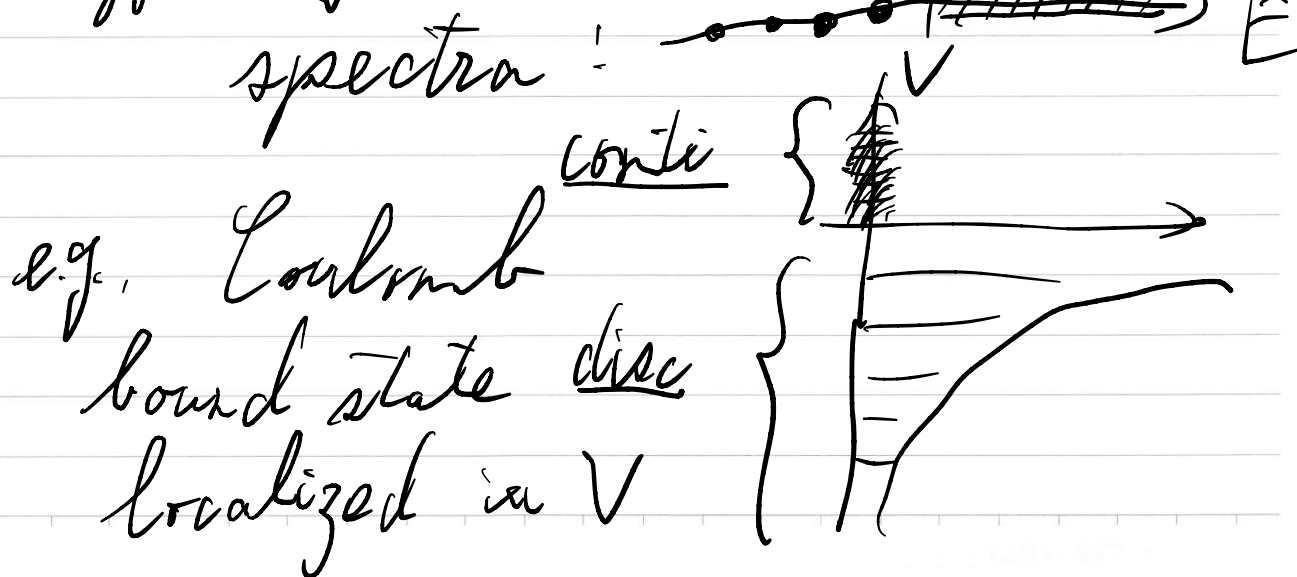
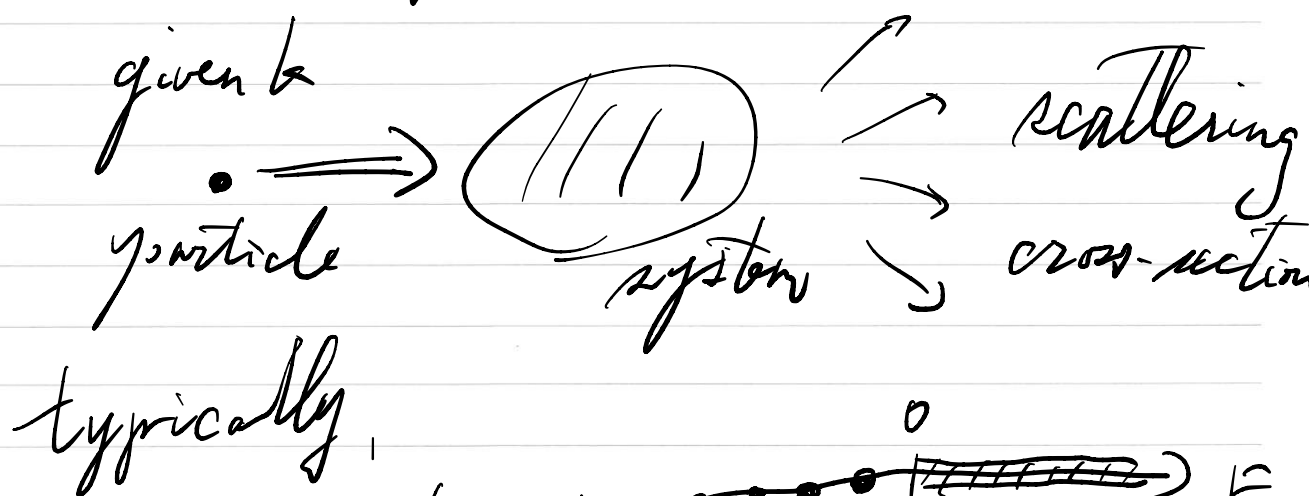
No. 1

Coffee meeting

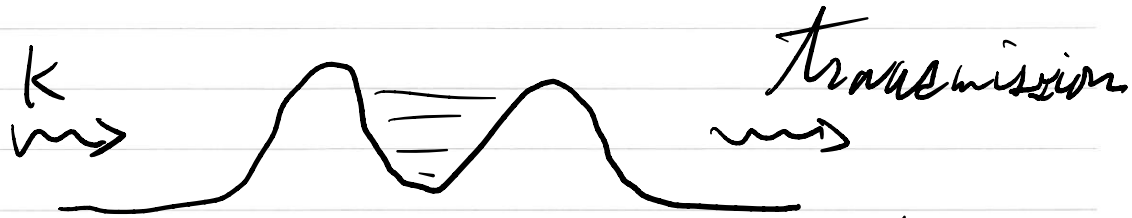
"Aspects of quantum resonance revisited"

Okuto Morikawa

② Quantum Mechanics (by von Neumann)



② Pseudo-bound state



"temporary bounded"

Unstable!! E ↑ $\left\{ \begin{array}{l} \text{disc, complex} \\ \Gamma : \text{decay width} \end{array} \right.$

• Resonance

① pole of S -matrix (trad.)

$\sim$ peak of scattering amplitude

② resolvent (math.)

$$\frac{1}{H - z} \quad z \in \mathbb{R}$$

analytic continuation $\leadsto$ sing.

③ physical state?

$$e^{ikx} = e^{ik_R x + k_I x} \rightarrow \infty$$

no wave func

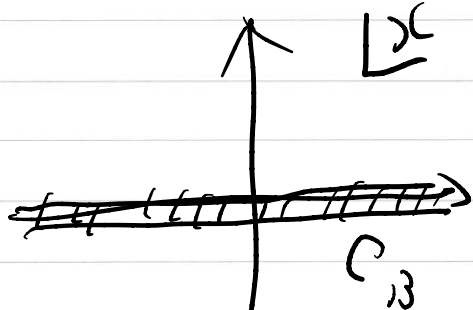
② Complex x (non-Hermitian QM)

At first, Hermitian QM $H^\dagger = H$

$$H\psi = E\psi \quad \text{eigenvalue}$$

$\uparrow$
 $\mathbb{R}$

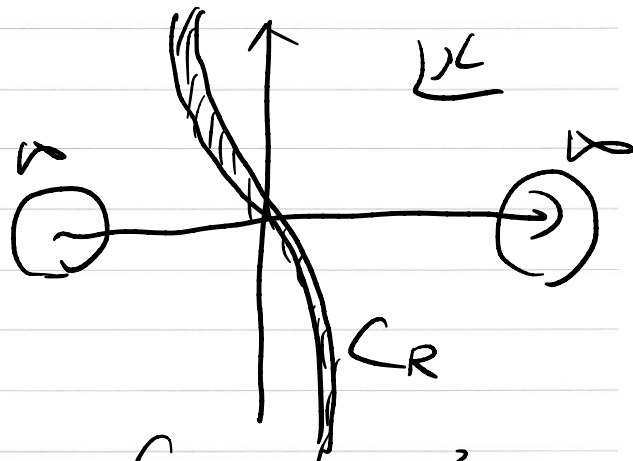
usual bound



$$\int_{-\infty}^{\infty} |\psi_B(x)|^2 < \infty$$

$$E \in \mathbb{R}$$

resonance



$$\int_{C_R} |\psi_R(x)|^2 < \infty$$

$$E \in \mathbb{C} !!$$

★ naively Hermitian,

but ill-defined on C_B at res.

Choice of $C_R \Leftrightarrow$ parameter space
of non-Hermitian QM

$$(H^\dagger = H, x \in \mathbb{C})$$

$$(H^\dagger \neq H, x \in \mathbb{R})$$